OOP: MCQs

Week1

Object-Oriented Programming MCQ Paper

Section A: Core Concepts of Object-Oriented Programming

1. What is an object in OOP?
a) A variable that holds primitive data
b) An instance of a class
c) A method to store data
d) A blueprint for a program

2. Which of the following is NOT a principle of OOP?
a) Abstraction
b) Encapsulation
c) Polymorphism
d) Compilation

3. Inheritance in OOP is used for:
a) Defining unrelated classes
b) Extending a class to reuse its code
c) Wrapping data and code into a single unit
d) Overloading methods

4. Which of these keywords is used to inherit a class in Java?
a) extend
b) inherit
c) implements
d) include

5. Polymorphism allows you to:
a) Write multiple methods with the same name but different parameters
b) Hide the internal details of an object
c) Create abstract classes
d) Automatically free up memory

Section B: Class Design and UML

6. What does UML stand for in software design?
a) Unified Mapping Language
b) Universal Modeling Layout
c) Unified Modeling Language
d) Universal Management Library

7. In UML, a private attribute is represented by which symbol?
a) +
b) -
c) *
d) #

8. What is the purpose of a constructor in a class?
a) To allocate memory for the class
b) To initialize objects
c) To define the class structure
d) To execute logic upon class destruction

9. Which of these is an example of an instance method?
a) static int sum()
b) private static void reset()
c) public double calculateInterest()
d) public Account()

Section C: Concurrency and Multithreading

10. What is the purpose of multithreading in Java?
a) To run multiple programs simultaneously
b) To enable a program to run multiple threads for efficient execution
c) To prevent errors during runtime
d) To create abstract methods

11. Which method is used to start a thread in Java?
a) run()
b) start()
c) execute()
d) begin()

12. What is the state of a thread after `start()` is invoked?
a) Running
b) Ready to run

c) Waiting
d) Terminated

Section D: Defensive Programming and Testing

13. What is defensive programming?
a) Writing code that prevents other programs from accessing it
b) Writing code that anticipates potential errors and handles them gracefully
c) Writing minimalistic code
d) Writing code optimized for multi-threading

14. Unit testing in Java is primarily performed using which framework?
a) JUnit
b) TestNG
c) Mockito
d) All of the above

Section E: Design Patterns

15. Which of these is NOT a type of design pattern?
a) Structural
b) Behavioral
c) Computational
d) Creational

16. The Singleton design pattern is used to:
a) Create multiple instances of a class
b) Ensure only one instance of a class exists
c) Protect the data in a class
d) Hide implementation details

Section F: Graphical User Interfaces

17. Which Java package is commonly used for GUI development?
a) java.net
b) javax.swing
c) java.io
d) java.util

18. In Java Swing, which method is used to make a frame visible?
a) setVisible(true)
b) show()
c) display()
d) activate()

Section G: Miscellaneous

19. What is the purpose of the `toString()` method in Java?
a) To compare two strings
b) To create a new string
c) To return a string representation of an object
d) To convert a string to a number

20. What is garbage collection in Java?
a) Manually deleting unused objects
b) Reclaiming memory used by unreferenced objects
c) A way to allocate memory for objects
d) A way to save objects to a file

Week2

Multiple-Choice Questions (MCQ) Paper: 5COSC019W – Object-Oriented Programming

Section A: Static Methods and Variables

1. **Which of the following is true about static variables in Java?**
    - a) They are associated with an instance of the class.
    - b) They are declared within methods only.
    - c) They are shared by all instances of the class.
    - d) They cannot be accessed without creating an object.

    Answer: c

2. **Static methods can access:**
    - a) Only static variables and methods.
    - b) Only dynamic variables and methods.
    - c) Both static and dynamic variables and methods.
    - d) Neither static nor dynamic variables.

Answer: a

3. **What is the correct way to access a static variable named count in a class Counter?**
    - a) Counter.count
    - b) Counter.getCount()
    - c) new Counter().count
    - d) Counter->count

Answer: a

4. **What will be the output of the following code?**

```
public class Demo {
    static int count = 0;
    public Demo() {
        count++;
    }
    public static int getCount() {
        return count;
    }
}
Demo d1 = new Demo();
Demo d2 = new Demo();
System.out.println(Demo.getCount());
```

    - a) 0
    - b) 1
    - c) 2
    - d) Compilation Error

Answer: c

Section B: Objects and Classes

5. **What is a constructor in Java?**
    - a) A method that returns the name of a class.
    - b) A special method used to initialize objects.
    - c) A method that is called only once during the program.
    - d) A static method used to call other methods.

Answer: b

6. **Given the following class, which line demonstrates object creation?**

```
public class Student {
    private String name;
    public Student(String name) {
        this.name = name;
    }
```

}

- a) Student.setName("John");
- b) new Student();
- c) Student student = new Student("John");
- d) student.Student("John");

Answer: c

12. **Which keyword is used to restrict access to variables within a class?**
    - a) static
    - b) private
    - c) public
    - d) final

    Answer: b

13. **What is the correct syntax to call a method `getName()` for an object `person`?**
    - a) person::getName();
    - b) person.getName();
    - c) getName(person);
    - d) getName::person();

    Answer: b

Section C: Static vs Non-Static Context

9. **Which of the following is NOT allowed in a static method?**
    - a) Accessing static variables
    - b) Calling static methods
    - c) Modifying dynamic variables directly
    - d) Returning static values

    Answer: c

10. **What will happen if a non-static variable is accessed in the static `main` method without an object?**
    - a) The program will run successfully.
    - b) Compilation error.
    - c) The variable will be initialized to null.
    - d) Runtime error.

    Answer: b

11. **Which of the following keywords can be used to create an instance of a class within the static `main` method?**
    - a) static
    - b) new
    - c) final
    - d) instance

    Answer: b

Section D: Java Predefined Classes

12. **Which package is automatically imported in every Java program?**
    - a) java.io
    - b) java.util
    - c) java.text
    - d) java.lang

    Answer: d

13. **What does the `length()` method in the `String` class return?**
    - a) The capacity of the string.
    - b) The memory allocated for the string.
    - c) The number of characters in the string.
    - d) The hash code of the string.

    Answer: c

14. **What is the correct syntax to use the `sqrt` method from the `Math` class?**
    - a) Math.sqrt(value);
    - b) Math.value.sqrt();
    - c) sqrt(value);
    - d) value.sqrt();

    Answer: a

Section E: UML Diagrams

15. **In UML, what does a solid line between two classes represent?**
    - a) Inheritance
    - b) Association
    - c) Aggregation
    - d) Dependency

    Answer: b

16. **What is the correct UML visibility symbol for a private field?**
    - a) +
    - b) -
    - c) #
    - d) ~

    Answer: b

17. **Which of the following is a composition relationship in UML?**
    - a) Student --- School
    - b) Leg --- Person
    - c) Course --- Faculty
    - d) Bank --- Customer

    Answer: b

18. **What does a use case diagram primarily describe?**
    - a) Relationships between objects
    - b) System architecture
    - c) External interactions with the system
    - d) Sequence of method calls

    Answer: c

Section F: Miscellaneous

19. **Which of the following statements about static variables is true?**
    - a) They are initialized by constructors.
    - b) They belong to the instance of the class.
    - c) They are shared across all instances of the class.
    - d) They must be declared as final.

    Answer: c

20. **Which keyword is used to declare a class as belonging to a specific package?**
    - a) package
    - b) import
    - c) public
    - d) static

    Answer: a

Week 3

MCQs for Object-Oriented Programming (5COSC019W)

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1. Inheritance

Q: What is the relationship between a subclass and its superclass?

- A. A subclass contains all private variables of the superclass
- B. A subclass inherits variables and methods from the superclass, except private ones
- C. A superclass inherits methods from its subclass
- D. A subclass cannot override methods of the superclass

Answer: B

Q: What keyword is used to call a superclass's constructor in Java?

- A. this
- B. super
- C. base
- D. parent

Answer: B

Q: Which of the following is NOT true about inheritance?

- A. It promotes code reuse
- B. Subclasses can override methods of the superclass
- C. A subclass can inherit constructors from its superclass
- D. It creates a hierarchical relationship between classes

Answer: C

---

2. Substitution Principle and Polymorphism

Q: According to the Substitution Principle, which statement is valid?

- A. An instance of a subclass cannot replace an instance of its superclass
- B. A superclass can replace an instance of its subclass
- C. An instance of a subclass can replace an instance of its superclass

- D. Subclasses and superclasses are unrelated

Answer: C

Q: Which of the following demonstrates polymorphism?

- A. A method with the same name but different parameters
- B. Using the same method call to operate on different object types
- C. A superclass calling a subclass's method
- D. Overloading a constructor

Answer: B

Q: Which is NOT true about polymorphism?

- A. It allows the same interface to work with different types
- B. It requires objects to have a common superclass
- C. It depends on dynamic method binding
- D. It resolves method calls during compile time

Answer: D

---

3. Classes and Objects

Q: Which of the following is true about instance variables?

- A. They are shared among all objects of the class
- B. They belong to specific objects created from the class
- C. They cannot be initialized in constructors
- D. They are accessible only within their class

Answer: B

Q: How do you create an object in Java?

- A. Using the new keyword and a constructor
- B. Using a static method
- C. Declaring a variable without initialization
- D. Using the super keyword

Answer: A

Q: Which modifier ensures a variable is accessible only within its class?

- A. public
- B. protected
- C. private
- D. static

Answer: C

---

4. Static vs. Instance Contexts

Q: What is the main difference between static and instance methods?

- A. Static methods can be overridden
- B. Instance methods belong to the class, static methods to objects
- C. Instance methods require an object to be invoked, static methods do not
- D. Static methods can access instance variables directly

Answer: C

Q: What does the keyword “static” indicate in Java?

- A. The method can access instance variables
- B. The method belongs to the class, not to any object
- C. The method can only be called within its class
- D. The method is private by default

Answer: B

---

5. Method Overriding and Dynamic Binding

Q: Which condition must be met for a method to be overridden?

- A. The method must be static
- B. The method must have the same name and parameters as the superclass method
- C. The method must be private
- D. The method must have a different return type

Answer: B

Q: What is dynamic binding in Java?

- A. Determining method calls at compile time
- B. Determining method calls at runtime
- C. Assigning variables to objects
- D. Overloading a method

Answer: B

Q: Which statement is true about dynamic binding?

- A. It works only with static methods
- B. It allows the program to decide which method to call during execution
- C. It is resolved during compilation
- D. It applies only to overloaded methods

Answer: B

---

6. Access Modifiers and Protected Variables

Q: What is the scope of a protected variable?

- A. Only within the same class
- B. Within the same class and subclasses
- C. Within the same package only
- D. Accessible from anywhere

Answer: B

Q: Which is a valid reason for using the protected modifier?

- A. To allow access to variables in unrelated classes
- B. To allow controlled access to variables in subclasses
- C. To prevent access from subclasses
- D. To restrict access within the package

Answer: B

---

7. Exercises

Q: Given the following code, what will be the result of executing emp.getName()?

```
public class Person {
   private String name;
   public Person(String name) { this.name = name; }
   public String getName() { return name; }
}

public class Employee extends Person {
   private double salary;
   public Employee(String name, double salary) {
      super(name);
      this.salary = salary;
   }
}

Employee emp = new Employee("John", 50000);
```

- A. John
- B. null
- C. 50000
- D. Compilation error

Answer: A

Q: What happens if you try to override a private method in a subclass?

- A. The method will override the superclass's method
- B. The subclass will inherit the method
- C. The subclass will define a new method unrelated to the superclass's method
- D. Compilation error

Answer: C

---

8. Instanceof and Casting

Q: What does the following code return?

```
Animal a = new Cat();
if (a instanceof Cat) {
   System.out.println("a is a Cat");
}
```

- A. Compilation error
- B. Runtime error

- C. "a is a Cat"
- D. Nothing

Answer: C

Q: When is casting object references required?

- A. When accessing methods of a superclass
- B. When calling private methods
- C. When accessing methods or fields of a subclass
- D. When creating a subclass object

Answer: C

Week 04

MCQs for Object-Oriented Programming (5COSC019W)

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1. Inheritance

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- A. A subclass contains all private variables of the superclass
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- C. A superclass inherits methods from its subclass
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- B. Using a static method
- C. Declaring a variable without initialization
- D. Using the super keyword

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- B. protected
- C. private
- D. static

Answer: C

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7. Exercises

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   public String getName() { return name; }
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   private double salary;
   public Employee(String name, double salary) {
      super(name);
      this.salary = salary;
   }
}

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```

- A. John
- B. null
- C. 50000
- D. Compilation error

Answer: A

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- A. The method will override the superclass's method
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Answer: C

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if (a instanceof Cat) {
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```

- A. Compilation error
- B. Runtime error
- C. "a is a Cat"
- D. Nothing

Answer: C

Q: When is casting object references required?

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- B. When calling private methods
- C. When accessing methods or fields of a subclass
- D. When creating a subclass object

Answer: C

Week 05

5COSC001W – Object-Oriented Programming MCQ Assessment

What We Learnt So Far

1. A real-life scenario can be modeled using:
    - A. Functions
    - B. Objects
    - C. Variables
    - D. Loops
2. Which statement is true about classes?
    - A. A class is a blueprint for objects.
    - B. A class is an instance of an object.

- C. Classes and objects are unrelated.
- D. Classes cannot have methods.

Object-Oriented Principles

3. What is encapsulation in Object-Oriented Programming?
   - A. Inheriting properties from a parent class.
   - B. Hiding the implementation details and exposing only the required functionalities.
   - C. Using the same method name for different functionalities.
   - D. Restricting the creation of objects from a class.
4. Inheritance allows:
   - A. A class to use methods and properties of another class.
   - B. Multiple classes to combine into one class.
   - C. A method to perform multiple operations.
   - D. Two classes to have identical names.
5. Polymorphism enables:
   - A. Overloading of methods.
   - B. Objects to take multiple forms.
   - C. Variables to store multiple types.
   - D. Only primitive data types in a class.

Summary Topics

6. What is true about the Java Object class?
   - A. It is an abstract class.
   - B. It is the root class of all Java classes.
   - C. It cannot be overridden.
   - D. It does not have any methods.
7. Which of these is a method of the Object class?
   - A. print()
   - B. equals()
   - C. append()
   - D. execute()
8. Why is the `toString()` method important?
   - A. It generates unique hashes for objects.
   - B. It provides a textual representation of an object.
   - C. It terminates the program.
   - D. It initializes object values.

Final Keyword

9. What does the `final` keyword do when used with a variable?
   - A. Prevents inheritance.

- B. Makes the variable immutable.
- C. Allows the variable to be null.
- D. Enables runtime changes.

10. Which of the following is true about a final class?
    - A. It can be subclassed.
    - B. It can be instantiated multiple times.
    - C. It prevents inheritance.
    - D. It must implement an interface.

Abstract Classes and Interfaces

11. What happens when a class contains an abstract method?
    - A. The class must be declared abstract.
    - B. The class cannot have any constructors.
    - C. The class must implement all methods of the Object class.
    - D. The class becomes a final class.
12. Why can't abstract classes be instantiated?
    - A. They contain unimplemented methods.
    - B. They don't have constructors.
    - C. They are immutable.
    - D. They don't allow inheritance.
13. Which is correct about interfaces?
    - A. They cannot have constants.
    - B. They can contain abstract methods only.
    - C. A class can implement multiple interfaces.
    - D. Interfaces are instantiated like classes.

Design Patterns

14. What is a design pattern in programming?
    - A. A concrete implementation of an algorithm.
    - B. A reusable solution to common design problems.
    - C. A data structure used to model objects.
    - D. A low-level programming construct.
15. Which element is NOT part of a design pattern?
    - A. Name
    - B. Problem
    - C. Solution
    - D. Algorithm Complexity
16. What does the Template Method Pattern do?
    - A. Allows multiple inheritance in Java.
    - B. Defines the skeleton of an algorithm while deferring steps to subclasses.
    - C. Prevents method overriding in subclasses.
    - D. Encapsulates multiple objects into one.

17. In the context of the Template Method Pattern, why is the template method marked as `final`?
    - A. To ensure it cannot be overridden.
    - B. To allow runtime changes.
    - C. To make it abstract.
    - D. To allow it to have no implementation.

General Questions

18. What does the UML stereotype “<>” represent?
    - A. A class that cannot have attributes.
    - B. A class containing abstract methods.
    - C. A class that cannot be extended.
    - D. A class that is immutable.
19. How does Java handle multiple inheritance?
    - A. Through abstract classes only.
    - B. By using interfaces.
    - C. Through the Object class.
    - D. Java does not support multiple inheritance.
20. Why might you choose to use an interface over an abstract class?
    - A. To allow a class to implement multiple behaviors.
    - B. To prevent instantiation.
    - C. To provide default method implementations.
    - D. To reduce runtime errors.

This MCQ paper evaluates your understanding of Object-Oriented Programming principles and related Java concepts. Good luck!

Week 07

5COSC001W – Object Oriented Programming

Multiple Choice Questions

Collections and Data Structures

1. Which of the following is true about collections in Java? a. Collections are always unordered and do not allow duplicates. b. All collections maintain an ordering of elements. c. Collections can contain duplicate elements depending on the implementation. d. Collections cannot be used with primitive types.
2. Which class is NOT a part of the Java Collections Framework? a. ArrayList b. HashMap c. PriorityQueue d. Hashtable

Arrays

3. What is the output of the following code?

```
int[] scores = {77, 88, 85, 41};
System.out.println(scores[2]);
```

a. 77 b. 85 c. 88 d. 41

6. Which of the following statements about arrays in Java is correct? a. Arrays can hold elements of different data types. b. Arrays must be manually resized by the programmer. c. Arrays are zero-indexed. d. Arrays automatically handle duplicate values.
7. What exception is thrown if an array index is accessed out of bounds? a. NullPointerException b. ArrayIndexOutOfBoundsException c. IndexOutOfRangeException d. IllegalArgumentException

List, Queue, Map

6. Which method is used to insert an element at a specific position in an ArrayList? a. addElement(index, value) b. insert(index, value) c. add(index, value) d. set(index, value)
7. Which of the following is NOT true about a Queue? a. A queue operates in a FIFO manner. b. Elements can be added at both ends of the queue. c. Elements are removed from the front of the queue. d. Queue operations include enqueue and dequeue.
8. Which class provides an implementation of the Map interface? a. HashMap b. ArrayList c. PriorityQueue d. LinkedList

Searching and Sorting

9. What is the purpose of the Arrays.binarySearch() method? a. To sort an array in ascending order. b. To search for a value in an unsorted array. c. To search for a value in a sorted array. d. To check if an array contains duplicate values.
10. Which of the following methods is used to sort an ArrayList? a. Arrays.sort() b. Collections.sort() c. ArrayList.sort() d. List.sort()

Linked List

11. Which of the following is NOT a characteristic of a linked list? a. Each element points to the next element. b. Linked lists can dynamically grow and shrink in size. c. Elements are stored in contiguous memory locations. d. Deleting an element from a linked list is faster than from an array.
12. What is the time complexity of accessing an element in a linked list? a. O(1) b. O(log n) c. O(n) d. O(n log n)

Arrays as Parameters

13. When an array is passed as a parameter to a method in Java, which of the following is true? a. A copy of the array is passed. b. The reference to the array is passed. c. Both the reference and the array elements are passed. d. Only the first element of the array is passed.
14. What is the output of the following code?

```
public static void doubleValues(int[] x) {
    for (int i = 0; i < x.length; i++) {
        x[i] *= 2;
    }
}

public static void main(String[] args) {
    int[] values = {1, 2, 3};
    doubleValues(values);
    System.out.println(values[1]);
}
```

a. 2 b. 3 c. 4 d. 6

Collections Methods

15. Which method in ArrayList removes an element by its value? a. remove(int index) b. remove(Object value) c. removeElement(Object value) d. delete(Object value)
16. What does the size() method of an ArrayList return? a. The maximum capacity of the ArrayList. b. The current number of elements in the ArrayList. c. The index of the last element in the ArrayList. d. The number of elements the ArrayList can still hold.

General

17. Which of the following is a key difference between an ArrayList and a LinkedList? a. ArrayList uses more memory than LinkedList. b. LinkedList provides constant-time positional access. c. ArrayList allows duplicate elements, while LinkedList does not. d. LinkedList is faster for inserting and deleting elements in the middle.
18. Which of the following statements about HashMap is false? a. A HashMap allows null keys and null values. b. The keys in a HashMap are unique. c. A HashMap maintains the order of insertion. d. A HashMap provides constant-time performance for basic operations.
19. What is the output of the following code?

```
Map<String, Integer> map = new HashMap<>();
map.put("A", 1);
map.put("B", 2);
map.put("C", 3);
System.out.println(map.size());
```

a. 1 b. 2 c. 3 d. 4

25. Which of the following operations is NOT supported by a Set? a. Adding elements b. Removing elements c. Retrieving an element by index d. Checking if an element exists

5COSC019W – Object Oriented Programming: GUI Programming MCQ Test

Section 1: Components, Layouts, and Events

1. What are the primary elements of GUI programming?
    - a) Frames, Panels, Tables
    - b) Components, Layouts, Events
    - c) Buttons, Text Fields, Labels
    - d) Listeners, Models, Views

    Answer: b) Components, Layouts, Events
2. What is the purpose of a layout in GUI programming?
    - a) To define the color of components
    - b) To control the positioning of components within a container
    - c) To set the behavior of event listeners
    - d) To draw components directly using the OS platform **Answer**: b) To control the positioning of components within a container
3. What is an event in the context of GUI programming?
    - a) A visual component appearing on the screen
    - b) A method to control component layout
    - c) A response to user actions
    - d) A class for creating frames **Answer**: c) A response to user actions

Section 2: AWT vs. Swing

4. Which package contains the Swing components?
    - a) java.swing
    - b) java.awt.swing
    - c) javax.swing
    - d) javax.awt **Answer**: c) javax.swing
5. Why are Swing components considered “lightweight”?
    - a) They depend on peer architecture
    - b) They draw themselves and do not rely on the OS platform
    - c) They use minimal memory
    - d) They are built on top of AWT **Answer**: b) They draw themselves and do not rely on the OS platform
6. Which of the following is a heavyweight component?
    - a) JButton
    - b) JLabel
    - c) Panel

    - d) JTextField **Answer**: c) Panel
7. Which method is required to make a JFrame visible?
    - a) show()
    - b) display()
    - c) setVisible(true)
    - d) render() **Answer**: c) setVisible(true)

Section 3: GUI Object Creation

8. What does the following code do? JLabel jlblName = new JLabel("Enter your name:");
    - a) Creates a text field
    - b) Creates a label with the text "Enter your name:"
    - c) Creates a button with the text "Enter your name:"
    - d) Creates a radio button with the text "Enter your name:" **Answer**: b) Creates a label with the text "Enter your name:"
9. Which class is used to create a combo box?
    - a) JComboBox
    - b) JList
    - c) JDropdown
    - d) JSelector **Answer**: a) JComboBox
10. What is the default layout for a JFrame?

- a) FlowLayout
- b) GridLayout
- c) BorderLayout
- d) CardLayout **Answer**: c) BorderLayout

Section 4: Layout Managers

11. Which layout manager places components sequentially from left to right?

- a) GridLayout
- b) BorderLayout
- c) FlowLayout
- d) CardLayout **Answer**: c) FlowLayout

12. How many components can be visible in a BorderLayout?

- a) Unlimited
- b) Maximum of five
- c) Two
- d) Six **Answer**: b) Maximum of five

13. What happens to components in a GridLayout when the frame is resized?

- a) They remain the same size
- b) They resize but lose the grid pattern
- c) They resize and maintain the grid pattern
- d) They overlap **Answer**: c) They resize and maintain the grid pattern

14. What is the purpose of using JPanel?

- a) To create frames
- b) To organize components and allow for multiple layouts within a frame
- c) To add event listeners
- d) To define the title of a window **Answer**: b) To organize components and allow for multiple layouts within a frame

Section 5: Colors and Fonts

15. Which method sets the background color of a component?

- a) setColor()
- b) setBackground()
- c) setForeground()
- d) setStyle() **Answer**: b) setBackground()

16. How would you create a font with bold style and size 16?

- a) `new Font("SansSerif", Font.BOLD, 16)`
- b) `new Font("SansSerif", Font.PLAIN, 16)`
- c) `new Font("SansSerif", Font.ITALIC, 16)`
- d) `new Font("SansSerif", Font.BOLD + Font.ITALIC, 16)` **Answer**: a) `new Font("SansSerif", Font.BOLD, 16)`

Section 6: Event Handling

17. What is the role of an event listener?

- a) To generate events
- b) To encapsulate information about an event
- c) To receive and process events
- d) To create GUI components **Answer**: c) To receive and process events

18. What happens when an event has no listener?

- a) It generates an exception
- b) It is ignored
- c) The program terminates
- d) It is logged for debugging **Answer**: b) It is ignored

19. What is the correct sequence of event handling?

- a) Event Source → Event Listener → Event Handler
- b) Event Handler → Event Source → Event Listener
- c) Event Listener → Event Source → Event Handler
- d) Event Source → Event Handler → Event Listener **Answer**: a) Event Source → Event Listener → Event Handler

20. Which interface must be implemented for handling button click events?

- a) KeyListener
- b) ActionListener
- c) MouseListener
- d) WindowListener **Answer**: b) ActionListener

5COSC019W – Object-Oriented Programming
MCQ Paper: GUI Programming in Java

Part 1: Components, Layouts, and Events

1. Which of the following is a visual object that appears on the screen?
   a) Container
   b) Component
   c) Event
   d) Panel
2. What does a layout control in Java GUI programming?
   a) Event handling
   b) Positioning of components within a container
   c) Creation of components
   d) Window resizing
3. What is the role of an event in GUI programming?
   a) To create components
   b) To draw graphics
   c) To respond to user actions
   d) To manage layouts

Part 2: Implementing GUI in Java

4. What is a key difference between AWT and Swing?
   a) AWT components rely on "peer architecture," while Swing components do not
   b) AWT is newer than Swing
   c) Swing components are heavyweight, whereas AWT components are lightweight
   d) Swing does not require the `javax.swing` package
5. Which Java GUI framework is designed for delivering rich internet applications (RIAs)?
   a) AWT
   b) Swing
   c) JavaFX
   d) Applet
6. What is the main advantage of JavaFX over older GUI technologies?
   a) It supports 2D and 3D graphics within a single library
   b) It requires the use of AWT methods
   c) It replaces Swing entirely
   d) It cannot access existing Java libraries

Part 3: Swing Components and Containers

7. Which of the following is NOT a Swing component?
   a) JButton
   b) JLabel
   c) JCheckbox
   d) Applet
8. What does the `JFrame` class represent in Swing?
   a) A container for other user interface components
   b) A label for displaying text
   c) A layout manager for organizing components
   d) A window with no title bar
9. What method is used to add a component to a JFrame's content pane?
   a) `setComponent()`
   b) `addComponent()`
   c) `add()`
   d) `setLayout()`

Part 4: Layout Managers

10. Which layout manager arranges components sequentially from left to right?
    a) GridLayout
    b) FlowLayout

c) BorderLayout
d) CardLayout

11. In a GridLayout, how are components arranged?
    a) Based on their size
    b) In rows and columns of equal size
    c) In five specific regions
    d) Sequentially, wrapping around
12. Which BorderLayout region dominates the sizing when a frame is resized?
    a) North
    b) South
    c) Center
    d) East

Part 5: Event Handling

13. What is an event listener in Java?
    a) A method to register GUI components
    b) An object that responds to events from a component
    c) A method that generates events
    d) A layout manager
14. What happens when there is no event listener for an event?
    a) The program crashes
    b) The event is ignored
    c) The event source stops functioning
    d) The application terminates
15. What are the three parts of the delegation event handling model?
    a) Source, Object, Component
    b) Source, Listener, Event Object
    c) Source, Panel, Method
    d) Object, Listener, Layout

Part 6: Advanced GUI Components and Features

16. How is a custom table model created for a JTable?
    a) By implementing the AbstractTableModel class
    b) By subclassing the TableModel interface
    c) By directly adding data to JTable
    d) By creating a new ArrayList
17. Which method is used to set the background color of a Swing component?
    a) setBackgroundColor()
    b) setColor()

c) setBackground()
d) setForeground()

18. Which class provides standard font names in Java?
    a) Color
    b) FontMetrics
    c) Font
    d) Dimension

Part 7: Miscellaneous

19. What does the Color class in Java use to define colors?
    a) CMYK model
    b) RGB model
    c) Grayscale model
    d) Alpha channel
20. What is the primary role of the MVC architecture in Swing?
    a) To simplify event handling
    b) To separate data, view, and user interaction
    c) To replace layout managers
    d) To manage colors and fonts

5COSC019W – Object Oriented Programming (Week 09)

MCQ Paper

Instructions:

- Each question carries one mark.
- Select the correct option by circling (A, B, C, or D).
- Time allowed: 20 minutes.
- No negative marking for wrong answers.

Questions:

1. What is the primary purpose of exception handling in object-oriented programming? A. To ensure the program compiles successfully B. To handle errors gracefully during execution C. To increase the execution speed of the program D. To replace all debug messages with exceptions

2. Which of the following keywords is used to handle exceptions in Java? A. error B. throw C. catch D. handle

3. Which of the following statements about the `finally` block is true? A. It is mandatory to use in every try-catch block. B. It executes only when an exception occurs. C. It is executed regardless of an exception occurring or not. D. It is used to declare exceptions.

4. Which of the following is an unchecked exception in Java? A. IOException B. ArithmeticException C. FileNotFoundException D. SQLException

5. In a try-catch block, which block is responsible for detecting and throwing exceptions? A. try B. catch C. throw D. finally

6. What will happen if a `throw` statement is executed outside of a try block? A. The program will terminate. B. The exception will be caught by default by the JVM. C. The program will go into an infinite loop. D. The program will compile but not execute.

7. What is the superclass of all exceptions in Java? A. Error B. Throwable C. RuntimeException D. Exception

8. Which of the following scenarios is best handled using a custom exception? A. Division by zero B. File not found C. Invalid user input for a specific business rule D. Array index out of bounds

9. Which block must directly follow a try block? A. another try block B. finally block C. catch block D. throws block

10. How many catch blocks can follow a single try block? A. Only one B. Exactly two C. None D. As many as required

11. What happens if an exception is thrown in a try block and no matching catch block is found? A. The program continues execution. B. The exception is ignored. C. The program terminates. D. The finally block executes and then the program continues.

12. Which of the following is true about checked exceptions? A. They can only occur during runtime. B. They must be declared in the method's throws clause or handled with a try-catch block. C. They are not detected at compile time. D. They represent serious errors that the program cannot recover from.

13. What is the output of the following code snippet?

```
try {
   int a = 5 / 0;
} catch (ArithmeticException e) {
   System.out.println("Arithmetic Exception");
} finally {
   System.out.println("Finally Executed");
}
```

A. Arithmetic Exception Finally Executed B. Arithmetic Exception C. Finally Executed D. Compilation Error

14. What is the purpose of `getMessage()` in the Throwable class? A. To display the stack trace of the exception B. To return a string containing a message about the exception C. To suppress the exception D. To determine the cause of the exception

15. Which type of exception must always be either caught or declared in the method's `throws` clause? A. RuntimeException B. Checked Exception C. Error D. Unchecked Exception

16. In which scenario is it appropriate to use `finally`? A. To ensure cleanup actions are performed regardless of what happens B. To replace catch blocks C. To handle unchecked exceptions D. To declare custom exceptions

17. Which of the following is an example of a runtime exception? A. NullPointerException B. IOException C. InterruptedException D. FileNotFoundException

18. How can a custom exception be created in Java? A. By extending the Throwable class directly B. By extending the Exception class C. By overriding the Throwable constructor D. By creating a new interface

19. Which of these is an example of black-box testing? A. Code walkthroughs B. Unit tests written with access to the code C. End-user testing of inputs and outputs D. Testing internal state changes

20. What is the main limitation of testing as a method for proving the correctness of a program? A. It is too time-consuming. B. It only removes all bugs in the program. C. It cannot show that the program will work in all possible scenarios. D. It does not support automation.

Answer Key: 1-B, 2-C, 3-C, 4-B, 5-A, 6-A, 7-B, 8-C, 9-C, 10-D, 11-C, 12-B, 13-A, 14-B, 15-B, 16-A, 17-A, 18-B, 19-C, 20-C

Question 1:

What is the purpose of exception handling in Object-Oriented Programming?

A. To make the code harder to debug.
B. To ensure the program exits immediately upon encountering an error.
C. To handle errors during execution in a controlled manner.
D. To eliminate the need for testing the code.

Answer:

Question 2:

Which of the following is NOT a common cause of program errors?

A. Memory leaks
B. Array out-of-bounds access
C. Converting an integer to a double
D. Divide by zero

Answer:

Question 3:

In Java, what is the keyword used to detect exceptions in a block of code?

A. catch
B. try
C. throw
D. finally

Answer:

Question 4:

Which of the following Java keywords is used to handle an exception after it is thrown?

A. finally
B. catch
C. throw
D. throws

Answer:

Question 5:

When does a finally block execute?

A. Only if an exception is thrown.
B. Only if an exception is not thrown.
C. Always, regardless of whether an exception is thrown or not.
D. Never, it is optional.

Answer:

Question 6:

What is the output of the following code snippet?

```
try {
   int a = 10 / 0;
} catch (ArithmeticException e) {
   System.out.println("Arithmetic Exception caught");
} finally {
   System.out.println("Finally block executed");
}
```

A. Arithmetic Exception caught
Finally block executed
B. Arithmetic Exception caught
C. Finally block executed
D. Compilation error

Answer:

Question 7:

Which of these classes is the superclass of all exceptions in Java?

A. Error
B. Throwable
C. Exception
D. RuntimeException

Answer:

Question 8:

What is a checked exception in Java?

A. An exception that is a subclass of RuntimeException.
B. An exception that must be declared in the method's `throws` clause if not caught.
C. An exception that does not need to be caught or declared.
D. An exception that can only occur during runtime.

Answer:

Question 9:

What will happen if you use multiple catch blocks, and the first catch block matches the superclass of the exception?

A. All catch blocks will execute sequentially.
B. The exception will be caught only by the first matching catch block.
C. The JVM will throw an error.
D. None of the catch blocks will execute.

Answer:

Question 10:

How can you create a custom exception in Java?

A. By defining a class that extends Throwable.
B. By defining a class that extends Exception.
C. By defining a class that extends RuntimeException.
D. Both B and C.

Answer:

Question 11:

Which statement is true about black-box testing?

A. It requires knowledge of the code implementation.
B. It focuses on testing inputs and outputs without knowing the code structure.

C. It is used exclusively for testing GUI applications.
D. It focuses on testing the internal states of an object.

Answer:

Question 12:

What is the main difference between black-box and white-box testing?

A. Black-box testing focuses on code paths, while white-box testing focuses on requirements.
B. Black-box testing focuses on inputs and outputs, while white-box testing focuses on internal code structure.
C. Black-box testing is manual, and white-box testing is automated.
D. Black-box testing tests small units, while white-box testing tests the entire system.

Answer:

Question 13:

Which built-in Java exception is thrown when you try to access an invalid index in an array?

A. IllegalArgumentException
B. ArrayIndexOutOfBoundsException
C. NullPointerException
D. IndexOutOfBoundsException

Answer:

Question 14:

What is the purpose of the `printStackTrace()` method in Java?

A. To terminate the program execution.
B. To print the cause of the exception and its stack trace.
C. To log the exception to a file automatically.
D. To rethrow the exception.

Answer:

Question 15:

What type of exception does the following code throw?

```
String s = null;
s.length();
```

A. NullPointerException
B. ArithmeticException
C. IllegalArgumentException
D. RuntimeException

Answer:

Object-Oriented Programming (OOP) MCQ Paper

1. **Which of the following statements about encapsulation is true?**
   - A) It hides data within a single class.
   - B) It exposes the internal implementation of a class.
   - C) It focuses only on reusability of code.
   - D) It makes all attributes of a class public.
     Answer: A) It hides data within a single class.
2. **What will the following code output?**

```
java
Copy code
class Example {
   String message = "OOP Concepts";
   void displayMessage() {
      System.out.println(message);
   }
}
public class Test {
   public static void main(String[] args) {
      Example e = new Example();
      e.message = "Encapsulation";
      e.displayMessage();
   }
}
```

   - A) OOP Concepts
   - B) Encapsulation
   - C) Compilation Error

- D) Runtime Exception

  Answer: B) Encapsulation

3. **Which of the following best describes abstraction?**
   - A) Hiding implementation details and exposing only functionality.
   - B) Making all methods private.
   - C) Encapsulating data into a single class.
   - D) Exposing internal logic for debugging.

     Answer: A) Hiding implementation details and exposing only functionality.
4. **What is the output of the following code?**

```
java
Copy code
abstract class Animal {
   abstract void sound();
}
class Dog extends Animal {
   void sound() {
      System.out.println("Woof");
   }
}
public class Main {
   public static void main(String[] args) {
      Animal a = new Dog();
      a.sound();
   }
}
```

   - A) Compilation error
   - B) Woof
   - C) Runtime exception
   - D) Abstract method cannot be instantiated

     Answer: B) Woof
5. **Consider the following code. What will it output?**

```
java
Copy code
class Parent {
   void display() {
      System.out.println("Parent");
   }
}
class Child extends Parent {
   void display() {
      System.out.println("Child");
   }
}
public class Main {
   public static void main(String[] args) {
      Parent p = new Child();
```

```
        p.display();
    }
}
```

- A) Parent
- B) Child
- C) Compilation Error
- D) Runtime Exception

Answer: B) Child

6. **Which of the following is not true about method overloading?**
   - A) It is a compile-time polymorphism.
   - B) It allows methods with the same name but different parameters.
   - C) It supports inheritance.
   - D) It cannot change the return type.

   Answer: C) It supports inheritance.
7. **What will happen if an interface method is declared as `private` in Java?**
   - A) It will work as expected.
   - B) Compilation error.
   - C) The method will not be accessible to implementing classes.
   - D) The method will be treated as default.

   Answer: B) Compilation error.
8. **What does the following code print?**

```
java
Copy code
interface A {
    default void show() {
        System.out.println("Interface A");
    }
}
class B implements A {
    public void show() {
        System.out.println("Class B");
    }
}
public class Main {
    public static void main(String[] args) {
        A obj = new B();
        obj.show();
    }
}
```

- A) Interface A
- B) Class B
- C) Compilation error
- D) Runtime exception

Answer: B) Class B

9. **Which keyword is used to rethrow an exception in Java?**
    - A) throw
    - B) throws
    - C) catch
    - D) finally

    Answer: A) throw
10. **What does the following code output?**

```
java
Copy code
public class Test {
   public static void main(String[] args) {
      try {
         int data = 10 / 0;
      } catch (ArithmeticException e) {
         System.out.println("Exception caught");
      } finally {
         System.out.println("Finally block executed");
      }
   }
}
```

    - A) Exception caught
    - B) Finally block executed
    - C) Exception caught Finally block executed
    - D) Runtime exception

    Answer: C) Exception caught Finally block executed
11. **What is the correct way to write data to a file in Java?**
    - A) Using Scanner class
    - B) Using FileOutputStream
    - C) Using Thread class
    - D) Using Console class

    Answer: B) Using FileOutputStream
12. **What does the following code snippet achieve?**

```
java
Copy code
BufferedReader br = new BufferedReader(new FileReader("input.txt"));
String line;
while ((line = br.readLine()) != null) {
   System.out.println(line);
}
br.close();
```

    - A) Writes data to the file
    - B) Appends data to the file
    - C) Reads data from the file and prints it

- D) Deletes the file
  Answer: C) Reads data from the file and prints it

xtended MCQ Questions

1. **Which of the following best describes polymorphism in Java?**
   - A) Writing multiple classes with the same name.
   - B) The ability of a class to have only one method.
   - C) The ability of different objects to respond uniquely to the same method.
   - D) Encapsulation of data within a single object.
     Answer: C) The ability of different objects to respond uniquely to the same method.
2. **What is the purpose of the `super` keyword in Java?**
   - A) To define a superclass.
   - B) To call the constructor or methods of the superclass.
   - C) To create a new instance of a superclass.
   - D) To access static members of a class.
     Answer: B) To call the constructor or methods of the superclass.
3. **Which of the following statements is true about abstract classes in Java?**
   - A) An abstract class cannot have concrete methods.
   - B) An abstract class can be instantiated directly.
   - C) An abstract class must have at least one abstract method.
   - D) An abstract class can have both abstract and concrete methods.
     Answer: D) An abstract class can have both abstract and concrete methods.
4. **What is the primary purpose of the `final` keyword in Java?**
   - A) To create a variable that can be reassigned.
   - B) To allow a class to be extended.
   - C) To declare constants, prevent inheritance, or method overriding.
   - D) To handle exceptions.
     Answer: C) To declare constants, prevent inheritance, or method overriding.
5. **What is the output of the following code?**

```
java
Copy code
class Test {
   public static void main(String[] args) {
      String str1 = "Java";
      String str2 = "Java";
      System.out.println(str1.equals(str2));
   }
}
```

   - A) true
   - B) false
   - C) Compilation error

- D) Runtime exception

  Answer: A) true

6. **Which of the following interfaces is not part of the Java Collections Framework?**
   - A) List
   - B) Map
   - C) Queue
   - D) HashTable

     Answer: D) HashTable
7. **What happens if the `hashCode()` method is not overridden when overriding the `equals()` method?**
   - A) Compilation error
   - B) Both methods will use their default implementations
   - C) Hash-based collections may behave unpredictably
   - D) It is mandatory to override both methods

     Answer: C) Hash-based collections may behave unpredictably
8. **Which of the following best describes the `synchronized` keyword in Java?**
   - A) It allows concurrent access to critical sections of code.
   - B) It ensures only one thread can execute a block of code at a time.
   - C) It stops threads from executing.
   - D) It makes variables immutable.

     Answer: B) It ensures only one thread can execute a block of code at a time.
9. **Which of the following best explains inheritance in Java?**
   - A) A way to achieve method overloading.
   - B) A mechanism to reuse code by deriving new classes from existing classes.
   - C) A way to implement encapsulation in code.
   - D) A technique to define static methods.

     Answer: B) A mechanism to reuse code by deriving new classes from existing classes.
10. **What is the purpose of the `static` keyword in Java?**
    - A) To create a variable that is shared among all instances of a class.
    - B) To define methods that can only be accessed from subclasses.
    - C) To create non-accessible methods.
    - D) To allow a method to override another method.

      Answer: A) To create a variable that is shared among all instances of a class.
11. **What does the `instanceof` keyword check in Java?**
    - A) Whether an object is an instance of a specific class or subclass.
    - B) Whether a method is defined in the current class.
    - C) Whether a variable is initialized.
    - D) Whether a class is abstract.

      Answer: A) Whether an object is an instance of a specific class or subclass.
12. **Which of the following statements about constructors is true?**
    - A) A constructor cannot be overloaded.
    - B) A constructor must have a `void` return type.
    - C) A constructor is called when an object is created.

- D) A class can have only one constructor.
  Answer: C) A constructor is called when an object is created.

13. **What will be the output of the following code?**

```
java
Copy code
public class Test {
   public static void main(String[] args) {
      String s = "Hello";
      s.concat(" World");
      System.out.println(s);
   }
}
```

- A) Hello World
- B) Hello
- C) Compilation error
- D) Runtime exception
  Answer: B) Hello

14. **Which of the following correctly describes the Java memory model?**
    - A) Stack memory is shared between threads.
    - B) Heap memory is used to store local variables.
    - C) Garbage collection manages memory by removing unreachable objects.
    - D) Static variables are stored in the stack.
      Answer: C) Garbage collection manages memory by removing unreachable objects.
15. **Which class is used for high-performance reading of characters, arrays, and lines?**
    - A) FileReader
    - B) BufferedReader
    - C) Scanner
    - D) InputStreamReader
      Answer: B) BufferedReader
16. **What is the primary difference between an interface and an abstract class in Java?**
    - A) Abstract classes support multiple inheritance, interfaces do not.
    - B) Interfaces can have constructors, abstract classes cannot.
    - C) An interface defines a contract, while an abstract class defines a partial implementation.
    - D) Abstract classes are used for utility methods only.
      Answer: C) An interface defines a contract, while an abstract class defines a partial implementation.

Concurrency MCQs

1. **Which of the following is true about the `volatile` keyword in Java?**
   - A) It prevents the creation of multiple threads.

- B) It guarantees visibility of changes to variables across threads.
- C) It allows only one thread to access a variable at a time.
- D) It enforces thread synchronization.

  Answer: B) It guarantees visibility of changes to variables across threads.

2. **What does the `wait()` method in Java do?**
   - A) It stops a thread permanently.
   - B) It pauses a thread until another thread notifies it.
   - C) It forces a thread to finish execution before others.
   - D) It initializes a thread.

     Answer: B) It pauses a thread until another thread notifies it.
3. **What is the purpose of the `ExecutorService` in Java?**
   - A) To create a new thread.
   - B) To manage and control a pool of threads.
   - C) To stop all threads immediately.
   - D) To prioritize threads.

     Answer: B) To manage and control a pool of threads.
4. **Which of the following is not a valid thread state in Java?**
   - A) NEW
   - B) READY
   - C) BLOCKED
   - D) TERMINATED

     Answer: B) READY
5. **What will the following code output?**

```
java
Copy code
public class Main {
   public static void main(String[] args) {
      Thread t = new Thread(() -> System.out.print("Hello "));
      t.start();
      System.out.print("World");
   }
}
```

   - A) Hello World
   - B) World Hello
   - C) Random output depending on thread scheduling
   - D) Compilation error

     Answer: C) Random output depending on thread scheduling

Collections MCQs

6. **Which of the following collections is ordered and allows duplicate elements?**

- A) HashSet
- B) ArrayList
- C) TreeMap
- D) LinkedHashSet

  Answer: B) ArrayList

7. **What is the time complexity of retrieving an element from a `HashMap` by its key?**
   - A) O(1) in the average case
   - B) O(n) in the average case
   - C) O(log n) in the average case
   - D) O(n^2) in the average case

     Answer: A) O(1) in the average case
8. **Which method is used to synchronize a collection explicitly?**
   - A) Collections.synchronize()
   - B) Collections.synchronizedCollection()
   - C) Collections.lock()
   - D) Collections.synchronized()

     Answer: B) Collections.synchronizedCollection()
9. **What will the following code output?**

```
java
Copy code
List<String> list = new ArrayList<>();
list.add("A");
list.add("B");
list.add(1, "C");
System.out.println(list);
```

   - A) [A, B, C]
   - B) [A, C, B]
   - C) [C, A, B]
   - D) [A, C]

     Answer: B) [A, C, B]
10. **Which collection type is best suited for maintaining unique elements in a natural order?**
    - A) ArrayList
    - B) HashSet
    - C) TreeSet
    - D) LinkedHashMap

      Answer: C) TreeSet

File Handling MCQs

11. **Which class is used to write primitive data types into a file in Java?**

- A) BufferedWriter
- B) FileOutputStream
- C) DataOutputStream
- D) ObjectOutputStream

  Answer: C) DataOutputStream

12. **What is the primary purpose of the `FileReader` class?**
    - A) To read bytes from a file
    - B) To read text from a file
    - C) To write text to a file
    - D) To delete files

      Answer: B) To read text from a file
13. **Which of the following is true about the `try-with-resources` statement?**
    - A) It automatically closes resources that implement the `AutoCloseable` interface.
    - B) It can only be used with `BufferedReader` and `FileReader`.
    - C) It prevents exceptions from being thrown.
    - D) It makes code execution faster.

      Answer: A) It automatically closes resources that implement the `AutoCloseable` interface.
14. **What will the following code output if the file `input.txt` contains the text "Hello World"?**

```
java
Copy code
BufferedReader br = new BufferedReader(new FileReader("input.txt"));
System.out.println(br.readLine());
br.close();
```

    - A) Hello
    - B) Hello World
    - C) Compilation error
    - D) IOException

      Answer: B) Hello World
15. **Which of the following methods is used to write text to a file in Java?**
    - A) writeText()
    - B) append()
    - C) write()
    - D) setText()

      Answer: C) write()
16. **What will happen if you attempt to read a non-existent file using `FileReader`?**
    - A) The file will be created.
    - B) It will throw a `FileNotFoundException`.
    - C) The program will terminate with a runtime error.
    - D) The method will return `null`.

      Answer: B) It will throw a `FileNotFoundException`.